Course Syllabus

1.	**Course ID**	2110651
2.	**Course Credit**	3 Credits
3.	**Course Title**	Digital Image Processing
4.	**Faculty / Department**	Engineering / Computer Engineering
5.	**Semester**	1
6.	**Academic year**	2025
7.	**Instructor**	Punnarai Siricharoen, PhD
8.	**Pre-requisite**	-
9.	**Course Type**	Approved Elective
10.	**Curriculum**	Computer Science
11.	**Degree**	Master's degree
12.	**Course date/time**	3 Hours Section 1: Thursday 9-12 (Section 1) Eng4 17-02 Section 5: Saturday 9-12 (Section 5) Eng4 18-16

13. Course Description

Visual perception, digitization and coding of images, converting pictures to discrete (digital) forms; image enhancement; image restoration including improving degraded low-contrast, blurred, or noisy pictures; image compression; data compression used in image processing; image segmentation referred to as first step in image analysis.

14. Course Information

14.1. Course Objectives

14.1.1. Be able to understand and describe fundamental image processing techniques

14.1.2. Be able to properly apply techniques for image analysis or image interpretation for the problems with intermediate complexity

14.1.3. Be able to design and implement image processing for image analysis and analyze the interpretation effectively

14.2. Content (see Tentative Schedule)

14.3. Teaching Methods

14.3.1 Onsite (Eng4, 17-02 (Thu), 18-16 (Sat))

14.3.2 myCourseVille course number : 2110651 password : 2110651 for course materials

14.3.3 Discord (announcement, Q&A)

14.4. Evaluation

14.4.1. Attendance (in-class exercise .5 per week + bonus) 5%

14.4.2. Homework / Exercises 10%

14.4.3. Project 25%

14.4.4. Midterm Exam 30%

14.4.5. Final Exam 30%

*** Your score will be reset to 0 point if any of your work is plagiarized ***

*** Late submission for assignment within 6 hours, receive a deduction of 10% of your mark,
within 24 hours, receive a deduction of 20% of your mark,
NO MARK for the submission after 24 hours due.

14.5 Grading

Score	Grade
>= 85	A
>= 75	B+
>= 70	B
>= 65	C+
>= 60	C
>= 55	D+
>= 50	D
< 50	F

15. Books

15.1. Books (Reference)

Rafael C. Gonzalez and Richard E. Woods, Digital Image Processing, Addison-Wesley

15.2. Additional materials: Related proceedings and Journals

Tentative Schedule

Week / Date Sec 1,5	Topic	Assignment HW (2 weeks due)
1 –7,9/8	Introduction to image processing and software preparation, Examples of fields using digital image processing	
2 – 14,16/8	Image Acquisition / Sampling and Quantization / Arithmetic Operations	
3 – 21,23/8	Histogram and Spatial Filtering	HW #1 Spatial Domain
4 – 28,30/8	Frequency Domain & Filtering in FD	
5 – 4,6/9	Wavelet and Multiresolution Processing	
6 – 11,13/9	Image Restoration	HW #2 Frequency & Research paper
7- 18,20/9	Image Compression	
8	Midterm exam (1 A4 paper, writing) Sat 27/9/2568 9-12	
9- 2,4/10	Fundamental Image Segmentation I	In-class discussion
10 – 9,11/10	Research discussion / Morphology Feature Extraction and Analysis	**Project Assignment**
11-16,18/10	Object Recognition (traditional /modern)	
12-25/10	Object Recognition II (23 Oct Holiday, watch video from 25 Oct) - **Hybrid**	HW #3 Segmentation + Image Augmentation
13-30,1/11	Image Segmentation II	Paper review presentation for project
14-6,8/11	Object Detection : Paper discussion & applications	
15-13,15/11	Image Generative Adversarial and other Imaging Applications – discussion (read and discuss)	
16-22/11	Invited Speaker (20 Nov no class, join a talk on 22 Nov)	
17	- **Final exam** Sat 6/12/2568 9-12 - **Project Presentation -** Thu 11/12/2568 18 – 21	